

AI Document Intelligence Report

Executive Summary

Overview

This sample report describes DocMind, a fictional document intelligence platform for portfolio demonstrations. The system ingests PDF documents, splits them into semantic chunks, embeds them with Hugging Face sentence-transformers, stores vectors in FAISS, and answers questions using retrieval-augmented generation.

Key Benefits

Grounded answers with source citations. Fast demo setup on free-tier infrastructure. Clear separation between conversational and document-specific questions. Portfolio-friendly architecture with FastAPI and Hugging Face Spaces.

Limitations

The demo does not replace legal, medical, or financial advice. OCR quality depends on the source PDF. Very long documents may be truncated by page limits. Retrieval may miss relevant passages if chunk boundaries split critical sentences.

Technologies Used

Backend: FastAPI and Uvicorn. RAG orchestration: LangChain community integrations. Embeddings: sentence-transformers/all-MiniLM-L6-v2. Vector store: FAISS. LLM: Groq ChatGroq with llama-3.3-70b-versatile. PDF parsing: PyPDFLoader.

Suggested Demo Questions

What is this document about? What are the main benefits? What are the limitations? What technologies are used? Summarize the document.

Conclusion

DocMind illustrates how a small, free-tier stack can deliver credible document Q&A for portfolios and prototypes.